Topic 4: Ontology-Grounded LLMs for Detecting Breast Cancer Misinformation in Patient Conversations

Problem & Hypothesis

Patients use online communities and LLMs for cancer information, but discussions contain misconceptions expressed indirectly—embedded in personal experiences or as questions with false presuppositions. The CancerMyth benchmark (585 questions) shows frontier LLMs frequently fail to identify these, and cautious prompting increases false positives.

Hypothesis: Explicit ontology grounding makes LLM-based misinformation detection more accurate, robust, and explainable than parametric knowledge or RAG alone.

Approach: Structured Claim Verification

The system (1) converts patient statements into structured claims (subject–relation–object triples), (2) grounds claims against a purpose-built Breast Cancer Misinformation Ontology (BCMO), and (3) verifies each against ontology constraints and retrieved evidence.

Example: "Chemotherapy won't work for HER2-positive" → `HER2+` → `incompatible_with` → `chemo` → flagged as misleading (both therapies are often combined).

Research Questions

RQ1: Does domain fine-tuning improve concept/relation extraction from patient text? **RQ2:** Does ontology grounding beat LLM-only for misinformation detection? **RQ3:** Does the ontology add value beyond standard RAG? **RQ4:** Does grounding improve robustness to paraphrases and implicit misinformation? **RQ5:** Does grounding produce more faithful explanations?

BCMO Ontology & Data Resources

Concepts: Cancer subtypes, biomarkers (ER/PR/HER2), treatments, side effects, risk factors, staging, metastasis, prognosis. **Relations:** causes, treats, contraindicated_for, increases_risk_of, has_side_effect, improves_survival. **Key feature:** Models negation, uncertainty, and relation strength (distinguishing `can reduce risk'' from`always prevents").

Data: Breastcancer.org (~362K posts), MedQuAD (47K QA pairs from NIH), CancerMyth (585+150 adversarial Qs), expert-annotated Reddit cancer misinformation dataset.

Related Work & Novelty Gap

- **FAITH (AAAI'26):** KG-grounded atomic claim verification for medical LLMs—uses general KGs, not cancer-specific.
- **ANCHOR-RE (2026):** Ontology-constrained biomedical relation extraction—targets scientific literature, not patient language.
- **CER (SIGIR'25):** Evidence retrieval + LLM reasoning for biomedical fact-checking—verifies news/web, not patient questions.
- **TrumorGPT / MedGraphRAG:** Graph-RAG for health fact-checking—no negation/uncertainty modeling.
- **OncoNet Ontology:** Cancer KG for biomarker discovery—no misinformation-specific relations.
- **GraphCheck / FactKG:** KG + GNN for multi-hop reasoning—general domain, no patient discourse focus.
- **Medical Sycophancy (2026):** Shows LLMs yield to patient misinformation pressure—no structured verification solution.

Our Unique Contribution

A purpose-built BCMO with negation/uncertainty/strength modeling, combined with structured claim decomposition from informal patient text—the first system bridging ontology-grounded verification with domain-specific breast cancer patient misinformation detection.